AI in Industry, weekend 2



Searching through representations

A story from ancient China

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



The court of King Xuan

The state of Qi, fourth century BC

King Xuan gathers scholars at his court. Good scholars are rare. Chunyu Kun, a witty advisor, brings him new ones.

How long would it take to find seven good scholars?

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



淳于髡一日而見七人於宣王。

In one day, Chunyu Kun presented seven men to King Xuan.

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain

King Xuan of Qi

reigned 319 to 301 BC

"Come. I have heard that one scholar in a thousand li is as if they stood shoulder to shoulder, and one sage in a hundred generations as if they came heel after heel. Now you present seven scholars in one morning: are scholars not plentiful, then?"

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer

Chunyu Kun

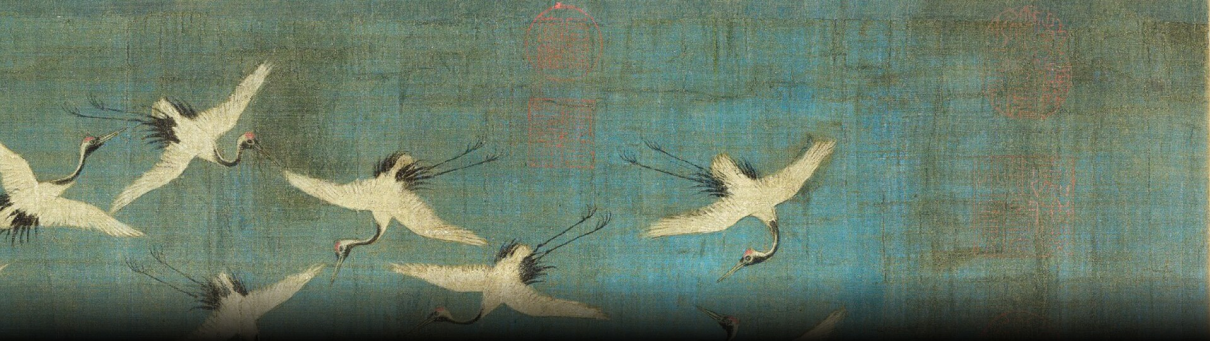
c. 386 to 310 BC

"Not so. Birds with the same wings live together; beasts with the same feet walk together. Look for chaihu and jiegeng in the marshes, and in generations you will not find one; go to the north slopes of the mountains 罩黍 and 梁父, and you load them by the cartful."

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer



Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain



"Things each have their kind, and I, Kun, am of the kind of the worthy. Your Majesty seeking scholars from me is like drawing water from a river or taking fire from a fire-drill. I will present more; why only seven scholars?"

Kun is himself a scholar, so scholars surround him.
To find more of a kind, look next to one you already know.

Zhanguo ce (Strategies of the Warring States), Qi 3.
English: lecturer

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain

Weekend 2

Representation learning

We organize data so that similar data points cluster together.

For each data point, we compute a representation, its embedding: a list of numbers that captures its most characteristic features.

Data points that belong together get similar embeddings.

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
Liaoning Provincial Museum. Wikimedia Commons, public domain

Weekend 2

Representation learning today

Each side gets an embedding, and the most similar are matched.

Etsy	a shopper's search with items for sale
Airbnb	a guest's search with homes
RAG systems	a user's prompt with a company's documents
LLMs	the text so far with the most suitable next token

Representation learning is one of the main drivers of AI today.

Emperor Huizong of Song, *Auspicious Cranes*, 1112, detail.
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